scale_abiotic_for_fit.R

August 12, 2026

Contents

scale_abiotic_for_fit 1

Index **3**

`scale_abiotic_for_fit` *Scale Abiotic Data for Model Fitting*

Description

Centres and scales abiotic predictor variables, records the scaling attributes for later back-transformation, and returns both the scaled data frame and the attributes as a named list.

By default, ‘age‘ is centred and scaled to unit standard deviation when age varies. Legacy centre-only scaling is available via ‘age_scale_mode = "center"‘. All other variables are both centred and scaled when more than one sample is present.

Usage

```
scale_abiotic_for_fit(data_abiotic_wide = NULL, age_scale_mode = "z_score")
```

Arguments

`data_abiotic_wide`
: A data frame in wide format as returned by ‘prepare_abiotic_for_fit()‘, containing real columns ‘dataset_name‘, ‘age‘, and one column per abiotic variable.

`age_scale_mode`
: Character scalar. ‘"z_score"‘ centres and scales ‘age‘ when age varies. ‘"center"‘ centres ‘age‘ only.

Details

Rows with any ‘NA‘ across the abiotic variables are silently dropped via ‘tidyr::drop_na()‘ before scaling. The returned ‘scale_attributes‘ list preserves the same structure as ‘attributes(scale(x))[-1]‘ (i.e., ‘dim‘ excluded).

Value

A named list with two elements:

‘data_abiotic_scaled‘ A data frame with row names in the format ‘"<dataset_name>__<age>"‘, a scaled ‘age‘ column, and all other scaled abiotic variable columns. Rows with any ‘NA‘ are dropped before scaling.

‘scale_attributes‘ A named list of ‘center‘ and ‘scale‘ attributes for each variable (including ‘age‘), which can be used to back-transform predictions.

See Also

[prepare_abiotic_for_fit()], [build_jsdm_fit_input()]

Index

scale_abiotic_for_fit, 1